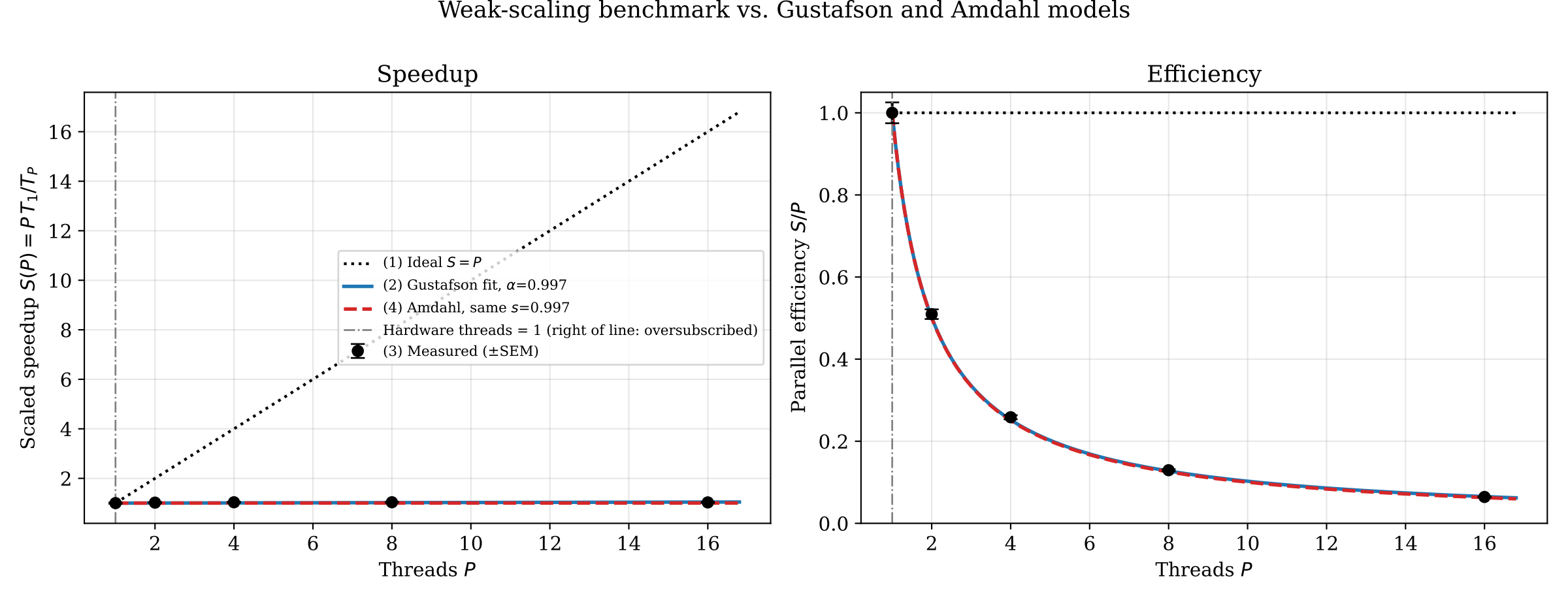
Weak-scaling benchmark vs. Gustafson and Amdahl models
Speedup
Scaled speedup $S(P) = P\,T_1/T_P$
Threads P
(1) Ideal $S = P$
(2) Gustafson fit, α=0.997
(4) Amdahl, same s=0.997
Hardware threads = 1 (right of line: oversubscribed)
(3) Measured (±SEM)
Efficiency
Parallel efficiency S/P
Threads P